Annexure 1: Scope of Work

Design & Supply of telescopic belt conveyor for the Empty Bag facility unloading

1. Current process Description

Received the Empty bags from different suppliers through different type of Trucks. The Empty Bag lot is unloaded in the Empty Bag storage shed Manually through carrying a 50 Kg lot on head. During this unloading activity, the Workers has to pass through a Ramp & travels approx. 15 to 50 Mtrs for stacking.

The frequency of receipt & unloading of Empty Bag Truck is Minimum one per day.

2. Requirement:

Design and supply of a telescopic belt conveyor which is capable of handling 20T/h load. Bag lot/bundle will be shifted in the telescopic belt conveyor manually which will deliver bundle in the available belt conveyor.

3. Technical Inputs:

Description	Requirement	Remarks
Service	Handling of Empty Bag	Yes
Weight of Empty Bag's lot	50 Kg	Yes
Capacity of belt conveyor	20 Ton	Yes
Dimension of Lot/bundle, (LxBxH) mm	1000 X 700X250	Ok
Belt & Telescopic Conveyor Speed	15-30 Mtr/Min	Yes
Belt Width	800mm	Yes
Bag Placement of Belt	Length wise	Yes
No of Collection Points	1 No	Yes
No of Delivery Points	1 Nos	Yes
Extension of telescopic conveyor	Total Length 14.5-meter, Fixed Length 5.5meter, Flexible length 3+3+3Meter.	Yes

Support	Hinged at one end and other end movable (up and down)	It must cover truck maximum loading point
Safety Devices		
Manual Stop switch at least three position	Required	Teknic make Contact rating-5Amp, IP65
Mechanical limit switch	1no-electric operated	Teknic -make Heavy duty 2NO+2NC Contact rating5Amp, IP65
Proximity Sensor for telescopic Conveyor	Capacitance type-2nos	2out of 1 logic with safe distance sensing at front
Others		
Belt Specifications	2 Ply 3.2 mm PVC	Yes
Material of structure	As per design	Approval to be taken, MOC of each component required in technical bidding
Driven Motor Make	CG/BBL , IE2	Yes
Panel for operation	As per design	Min-IP65, Ammeter, indication for each motion, with proper marking switch, with cannopy
Gear Box Make	3 HP [With Bonfiglioli Gearbox]	Yes
Power	All component should energy efficient	415Volt 50Hz 3-Phase
Duty	Intermittent	Dependent truck frequency, Min 2 truck per day
Clearance	Minimum clearance 100mm from platform	Truck platform height is 1350
Design	Design basis and calculation approval will be taken after getting order	will ensure for the minimum strength and power

**Preference will be given to minimum static (folded) length conveyor*.

4. Scope of Work

The Supplier's scope of work shall be the complete Detailed Engineering, Procurement of Materials, Manufacturing, testing & supply of telescopic Conveyor with required accessories.

The scope of supply includes but not limited to the followings-

4.1 Design of telescopic Conveyor as per sheet & submission design & GA drawing for submission of same for approval.

4.2 Procurement of all material required for Telescopic conveyor.

4.3 Establishment of Quality Assurance Plan & submit for approval.
4.4 Procurement & Fabrication as per finalized Quality Assurance Plan
4.5 Trial of complete Telescopic Conveyor at Supplier works
4.6 Supply of complete telescopic conveyor at site.
4.7 Submission of all the As-built drawings, Data sheets & QAP adherence with necessary documents for final records.
4.8 Submission of Project Schedule, Monthly Progress Report.

Note: Supplier is suggested to Maximum utilization of modern technologies after approval.

5. Special requirements:

5.1 **Pulleys:**

5.1.1 All drum pulleys (Head & Tail Pulley) are to be provided with XT hub & bushing/ shrink fit instead of conventional (only keyed hub) pulley.
5.1.2 Head Pulley shall have 12mm thk Natural Rubber lagging Diamond pattern groove, hot vulcanized method,
5.1.3 Tail Pulley shall have min 8mm thick Natural Rubber plain lagging, hot vulcanized method,
5.1.4 Pulley runout shall be within +/-0.5% of the nominal O.D (without lagging)
5.1.5 Pulley shall be statically balanced
5.1.6 Keyway as IS 2048
5.1.7 All weld is min, 6 mm fillet unless or otherwise mentioned
5.1.8 Shell plates shall have single longitudinal continuous weld
5.1.9 Shor hardness shall be 60°+/-5° on Scale A

5.2 **Idler Units**

5.2.1 Each roller for idlers shall consist of a pair of anti-friction bearings of ample capacity and shall have minimum double labyrinth seals for keeping the lubricant in and dirt out. The bearings and sealing devices shall be mounted in the housing with suitable rain cap to prevent water entering inside the housing. The bearing housing shall be welded on either side of the rollers with through shafts for proper load support.
5.2.2 The bearing housings shall be minimum 3 mm thick and shall be made of pressed steel, welded to the idler rollers simultaneously. The maximum co-efficient of friction for idler unit shall not exceed 0.015.
5.2.3 The carrying idler units shall be heavy duty suitable for continuous operation.
5.2.4 Return idlers shall be mounted on sturdy steel or malleable iron brackets bolted to the underside of stringers. Return idlers spacing near the head and tail pulleys shall be less than 2 meters.
5.2.5 The Sharp corner of Idler bracket shall be round off & having slots in place of bolt holes to align the belt during running conditions.
5.2.6 All carrying & return idlers of shall be rubber lagged.
5.2.7 All bearings and mounting should be **SKF make** only
5.2.8 A common greasing position for the all bearings to be provided

5.3 **Structural Details**

5.3.1 All structural steel material shall conform to IS 2062. All shop connections and field connections shall be welded or bolted type. Field connections in equipment supports and equipment platforms shall be with friction grip high strength bolts or welded connections.

5.3.2 Civil foundation detail and drawings to be provided for the Telescopic conveyor

5.4 **Motor Details**

1.1.1 Terminal box should be metallic.

1.1.2 Motor Insulation must be at least F class

1.1.3 Efficiency Class: IE2.

1.1.4 Motor Duty is continuous rated i.e. S1

1.1.5 IP class must be IP55 (Ingress Protection)

1.1.6 Motor Cooling fan should be metallic.

1.1.7 TEFC Motor, suitable up to 50° C ambient temp.

1.1.8 3 Phase; 415 V ± 10%; 50 Hz ± 5%

1.1.9 Motor Terminal box for New BC 16 should be at the top & the cable entry should be from walkway platform side. The Terminal box should be suitable foe 3Cx10 Sqmm.

1.1.10 For portable conveyors, Motor & Gear box should be parrel to the conveyor. Motor Terminal box should be according to Cable Entry form open area.

5.5 **Others**

5.5.1 The vibration limit shall be as per zones A and B of ISO 10816-1.

5.5.2 Noise levels produced by any rotating equipment individually and collectively shall not exceed 85 dB (A) measured at distance of 1.5 Mtrs from the source in any direction under any load condition.

5.5.3 Idlers shall be with grease packed sealed deep grove anti-seize type ball bearing. Bearings shall have multi labyrinth seals to protect from dust.

5.5.4 All the Fasteners shall be as per ASTM A 193 B7/ASTM A 194B8 with two washers.

2. Painting:

2.1 All CS external surfaces of shop fabricated equipment/item shall be painted as per the following.

2.2 Surface Preparation: Grit blasting to SA 2-1/2 Grade.

2.3 Total DFT of painting shall be 300µ (including primer, intermediate & finish paint) & the final coat shall be Red colour.

Note:

The paint specification shall be compatible with the external environment consisting of fine urea dust particles which is very corrosive and hygroscopic in nature.

3. Minimum Inspection Requirements:

3.1 **Idler Assembly**

3.1.1 TC verification of Raw materials (MS tubes, shafts, bearings, Brackets etc.).

3.1.2 Welding document (including NDT) review.

3.1.3 Final inspection of Brackets.

3.1.4 Round-off edges of Brackets.

3.2 **Idler Roller**

3.2.1 Dimensions of rollers, quality, and radial run out checks.

3.2.2 Shore hardness of rubber discs (for impact idlers).

3.2.3 Open inspection of bearing and rollers (at random).

3.3 **Idler with bracket**

3.3.1 Dimensional & troughing angle measurement of brackets.

3.3.2 Free movements of rollers.

3.3.3 Gap between roller edges.

3.3.4 Level of rollers.

3.3.5 Verification of thrust bearing, swiveling of self-aligning idlers about the pivot.

3.4 **Pulleys**

3.4.1 Raw material TC verification (Pulley shell, shaft, hubs, end discs, Plummer blocks, bearing etc.)

3.4.2 Welding document (including NDT) review.

3.4.3 Inspection as per drawing, dimensional measurements.

3.4.4 Lagging rubber examination and verification of hardness.

3.4.5 Static balancing after rubber lagging.

3.4.6 Verification of bearings.

3.4.7 Examination of hub and shaft fixing arrangement.

3.4.8 UT for shaft diameter>50mm.

3.5 **Gear Box:**

3.5.1 Gear Blanks, Shafts, Bearings, Pinions & Housing: MTC, 100% UT of forging,

3.5.2 Gears: Heat Treatment Chart Review, Hardness Check, Case depth for hardened and ground gears, PT on gear tooth

3.5.3 Performance test (No Load): Vibration Check, b) Noise level, c) Verification of Temperature Rise of Oil Bath d) Verification of speed ratio, e) Dimensional Inspection

3.6 **Structural Items:**

3.6.1 Raw material TC verification.

3.6.2 Welding document (including NDT) review.

3.6.3 Final inspection assembly- Visual, dimensional.

3.7 **Painting**

3.7.1 Surface Finish measurement

3.7.2 DFT measurement

3.8 **Electrical Motors**

3.8.1 Performance Test Certificate

4. Expected Completion Time

The delivery of above assignment shall be in a 2 months' time frame from the date of award of contract.

5. Applicable Engineering Codes & Standards:

It is the primary responsibility of vendor (Successful Supplier) to design & construct all items related to this contract in accordance with the latest industry standard & codes, including all supplements and addendums issued at time of the order. The applicable codes shall include but not limited to the followings.

IS 11592: Selection & Design of Belt Conveyor.
CEMA: Belt Conveyors for bulk materials.
IS 1891: Conveyor & Elevator Textile Belting.
IS 8531: Pulleys for Belt Conveyor
IS 8598: Idlers & Idlers sets for Belt Conveyor.
IS 9295: Steel Tubes for Idlers for Belt Conveyor.
International Journal of Engineering Trends and Technology (IJETT) – Volume 61 Number 2 – July 2018

6. Preservation, protection, Packing & Dispatch:

6.1 Cleaning, preservation, storage, packing and preparation for Loading/dispatch of the equipment, equipment components shall be performed according to the manufacturer's standard/good engineering practices.

6.2 All spares parts shall be packed separately in wooden packet to avoid damage in transport.

Performance Guarantees:

6.2.1 Telescopic Conveyor and belt feeder shall run centrally without any misalignment.

6.2.2 The vibration limit of conveyor drives as measured at center line of pulley bearings, motor bearings and gear box bearings shall be as per zones A and B of ISO 10816-1.

6.2.3 The noise level at a distance of 1.5 Mtrs from the drive of conveyor shall be less than 85 dB (A).

6.2.4 The vibrations in structural supports of conveyor structures shall be less than that specified

Enclosures

Concept layout for Empty Bag facility: Assessment of Telescopic Conveyor Height & length.